

Q1 - 25 June - Shift 2

If the mean deviation about the mean of the numbers 1, 2, 3, ..., n, where n is odd, is $\frac{5(n+1)}{n}$, then n is equal to _____

*Space for your notes:***Q2 - 26 June - Shift 1**

The mean of the numbers a, b, 8, 5, 10 is 6 and their variance is 6.8. If M is the mean deviation of the numbers about the mean, then 25 M is equal to:

Space for your notes:

- (A) 60 (B) 55 (C) 50 (D) 45

Q3 - 26 June - Shift 2

The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to :

Space for your notes:

- (A) 10 (B) 36 (C) 43 (D) 60

Q4 - 27 June - Shift 2

The mean and variance of the data 4, 5, 6, 6, 7, 8, x, y where $x < y$ are 6, and $\frac{9}{4}$ respectively. Then $x^4 + y^2$ is equal to

Space for your notes:

- (A) 162 (B) 320
(C) 674 (D) 420

Q5 - 28 June - Shift 1

The mean and standard deviation of 15 observations are found to be 8 and 3 respectively. On rechecking it was found that, in the observations, 20 was misread as 5. Then, the correct variance is equal to _____.

*Space for your notes:***Q6 - 29 June - Shift 1**

Let the mean and the variance of 5 observations x_1, x_2, x_3, x_4, x_5 be $\frac{24}{5}$ and $\frac{194}{25}$ respectively.

Space for your notes:

If the mean and variance of the first 4 observation are $\frac{7}{2}$ and a respectively, then $(4a + x_5)$ is equal to:

- (A) 13 (B) 15
(C) 17 (D) 18

Q7 - 29 June - Shift 2

The number of values of $a \in \mathbb{N}$ such that the variance of 3, 7, $12a$, $43 - a$ is a natural number is:

Space for your notes:

- (A) 0 (B) 2
(C) 5 (D) infinite

Answer Key

Q1 (21)

Q2 (A)

Q3 (C)

Q4 (B)

Q5 (17)

Q6 (B)

Q7 (A)

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Hints and Solutions

MathonGo

Q1 (21)Mean deviation about mean of first n naturalnumbers is $\frac{n^2 - 1}{4n}$ $\therefore n = 21$ **Q2 (A)**

$$\sigma^2 = \frac{\sum_{i=1}^5 (x_i - \bar{x})^2}{n}$$

Mean = 6

$$\frac{a + b + 8 + 5 + 10}{5} = 6$$

$$a + b = 7$$

$$b = 7 - a$$

$$6.8 = \frac{(a-6)^2 + (b-6)^2 + (8-6)^2 + (5-6)^2 + (10-6)^2}{5}$$

$$34 = (a-6)^2 + (7-a-6)^2 + 4 + 1 + 18$$

$$a^2 - 7a + 12 = 0 \Rightarrow a = 4 \text{ or } a = 3$$

$$a = 4 \quad a = 3$$

$$b = 3 \quad b = 4$$

$$M = \frac{\sum_{i=1}^5 |x_i - \bar{x}|}{n}$$

$$M = \frac{|a-6| + |b-6| + |8-6| + |5-6| + |10-6|}{5}$$

when $a = 3, b = 4$

$$M = \frac{3 + 2 + 2 + 1 + 4}{5}$$

$$M = \frac{12}{5}$$

$$25M = 25 \times \frac{12}{5} = 60$$

when $a = 4, b = 3$

$$M = \frac{2 + 3 + 2 + 1 + 7}{5}$$

$$M = \frac{12}{5}$$

Q3 (C)

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Hints and Solutions

MathonGo

No. of observations: - 50

$$\text{mean}(\bar{x}) = 15$$

Standard deviation (σ) = 2

Let incorrect observation is x_1 & correct observation is (x'_1)

$$\text{Given } x_1 + x'_1 = 70$$

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_{50}}{50} = 15 \text{ (given)}$$

$$\Rightarrow x_1 + x_2 + \dots + x_{50} = 750 \quad \dots \text{(i)}$$

Now

Mean of correct observation is 16

$$\frac{x'_1 + x_2 + \dots + x_{50}}{50} = 16$$

$$x'_1 + x_2 + x_3 + \dots + x_{50} = 16 \times 50 \quad \dots \text{(ii)}$$

eq. (ii) - eq. (i)

$$\Rightarrow x'_1 - x_1 = 16 \times 50 - 15 \times 50$$

$$x'_1 - x_1 = 50 \text{ \& } x_1 + x'_1 = 70$$

$$x'_1 = 60$$

$$x_1 = 10$$

$$\Rightarrow 4 = \frac{x_1^2 + x_2^2 + \dots + x_{50}^2}{50} - 15^2 \quad \dots \text{(iii)}$$

$$\Rightarrow \sigma^2 = \frac{x_1^2 + x_2^2 + \dots + x_{50}^2}{50} - 16^2 \quad \dots \text{(iv)}$$

from (iii)

$$\Rightarrow 4 = \frac{(10)^2}{50} + \frac{x_2^2 + x_3^2 + \dots + x_{50}^2}{50} - 225$$

$$\Rightarrow 4 = 2 - 225 + \frac{(x_2^2 + x_3^2 + \dots + x_{50}^2)}{50}$$

$$\Rightarrow 227 = \frac{(x_2^2 + x_3^2 + \dots + x_{50}^2)}{50}$$

From (iv)

$$\sigma^2 = \frac{(60)^2}{50} + \left(\frac{x_2^2 + x_3^2 + \dots + x_{50}^2}{50} \right) - (16)^2$$

$$\sigma^2 = \frac{60 \times 60}{50} + 227 - 256$$

$$\sigma^2 = 72 + 227 - 256$$

$$\sigma^2 = 43$$

Q4 (B)

Hints and Solutions

MathonGo

$$\text{mean } \bar{x} = \frac{4+5+6+6+7+8+x+y}{8} = 6$$

$$\Rightarrow x + y = 48 - 36 = 12$$

Variance

$$= \frac{1}{8} (16+25+36+36+49+64+x^2+y^2) - 36 = \frac{9}{4}$$

$$\Rightarrow x^2 + y^2 = 80$$

$$\therefore x = 4; y = 8$$

$$x^4 + y^2 = 256 + 64 = 320$$

Q5 (17)

We have

$$\text{Variance} = \frac{\sum_{r=1}^{15} x_r^2}{15} - \left(\frac{\sum_{r=1}^{15} x_r}{15} \right)^2$$

Now, as per information given in equation

$$\frac{\sum x_r^2}{15} - 8^2 = 3^2 \Rightarrow \sum x_r^2 = \log 5$$

$$\text{Now, the new } \sum x_r^2 = \log 5 - 5^2 + 20^2 = 1470$$

$$\text{And, new } \sum x_r = (15 \times 8) - 5 + (20) = 135$$

$$\therefore \text{Variance} = \frac{1470}{15} - \left(\frac{135}{15} \right)^2 = 98 - 81 = 17$$

Q6 (B)

$$\bar{x} = \frac{\sum x_i}{5} = \frac{24}{5} \Rightarrow \sum x_i = 24$$

$$\sigma^2 = \frac{\sum x_i^2}{5} - \left(\frac{24}{5}\right)^2 = \frac{194}{25}$$

$$\Rightarrow \sum x_i^2 = 154$$

$$x_1 + x_2 + x_3 + x_4 = 14$$

$$\Rightarrow x_5 = 10$$

$$\sigma^2 = \frac{x_1^2 + x_2^2 + x_3^2 + x_4^2}{4} - \frac{49}{4} = a$$

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = 4a + 49$$

$$x_5^2 = 154 - 4a - 49$$

$$\Rightarrow 100 = 105 - 4a \Rightarrow 4a = 5$$

$$4a + x_5 = 15$$

Q7 (A)

Mean = 13

$$\text{Variance} = \frac{9 + 49 + 144 + a^2 + (43 - a)^2}{5} - 13^2 \in \mathbb{N}$$

$$\Rightarrow \frac{2a^2 - a + 1}{5} \in \mathbb{N}$$

$$\Rightarrow 2a^2 - a + 1 - 5n = 0 \text{ must have solution as}$$

natural numbers

its $D = 40n - 7$ always has 3 at unit place

$\Rightarrow D$ can't be perfect square

So, a can't be integer.

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